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\\Research

Investigators - validated on FIT

Frederick

Potpourri

localizer_64ch_uncombined AAHead_Scout_64ch T1w_setter T1_MEMPRAGE_1.0mm_p4_vNav Minn_CMRR_2.3mm_S8_rest_6min CMRR_SEFieldMap_2.3mm_64sl_AP ABCD_T1w_MPR_vNav_setter ABCD_T1w_MPR_vNav ABCD_T2w_SPC_vNav_setter ABCD_T2w_SPC_vNav rfMRI REST ME PA XA60 cmrr_mbep2d_diff ep2d_bold_mgh ep2d_diff_mgh ep2d_se_sms_mgh ABCD_fMRI_rest_CMRR ABCD_fMRI_rest_MGH can_neuromelanin
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\localizer_64ch_uncombined

TA: 14 sec Coil Selection: Manual Voxel Size: 0.5×0.5×7.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	7.0 mm
TR	8.6 ms
TE	4.00 ms
Averages	2
Concatenations	3
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	8.6 ms
TE	4.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None

Contrast - Common

Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	7.0 mm
Base Resolution	256
Phase Resolution	91 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %

Geometry - Common

Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	7.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	91 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
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Inline - Subtraction

Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	4.00 ms
TR	8.6 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\AAHead_Scout_64ch

TA: 14 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head
Coil Elements	HC1-7;NC1

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FOV Read	260 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	1.37 ms
TR	3.15 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\T1w_setter

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H40.0
L	0.0 mm
P	0.0 mm
H	40.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	50.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
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Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	MPRAGE
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\T1_MEMPRAGE_1.0mm_p4_vNav

TA: 4:09 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	2nd Segment
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2200.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	HC1-7;NC1

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	4
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2200.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Contrast - Common

TR	2200.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. Preparation	Non-sel. IR
TI	1100 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	4
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	16.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	16.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	192 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	1100 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	7 deg
Measurements	1
Contrasts	4
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
TR	2200.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast

Sequence - Part 1

Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Reordering	Linear
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px
Bandwidth 4	650 Hz/Px
Echo Spacing	9.84 ms
Asymmetric Echo	Off
Turbo Factor	168

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav
Remeasure	20 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	RMS Only

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	207 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HC1-7

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FOV Read	207 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	207 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Log Signals	Off
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\CMRR_SEFieldMap_2.3mm_64sl_AP

TA: 8 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	207 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain
Coil Elements	HC1-7

Contrast - Common

TR	8000.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	207 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	207 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8000.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	On
EPI Factor	90

Sequence - Part 2

Introduction	Off
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Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ABCD_T1w_MPR_vNav_setter

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	8.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
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Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	MPRAGE
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ABCD_T1w_MPR_vNav

TA: 7:12 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
TE	2.90 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	2500.0 ms
TE	2.90 ms
Magn. Preparation	Non-sel. IR
TI	1070 ms
Flip Angle	8 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2500.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm
F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	5.000

Physio - Signal

1st Signal/Mode	None
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Physio - Signal

TR	2500.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI	1070 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.90 ms
TR	2500.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl_vn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	8.10 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav
Remeasure	24 TRs
Reacq. threshold	0.50
Feedback Delay	60 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	None

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ABCD_T2w_SPC_vNav_setter

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	13.0 ms
TE	6.10 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	BC

Contrast - Common

TR	13.0 ms
TE	6.10 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Water Excitation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	8.00 mm
TR	13.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	256 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	8.000

Physio - Signal

1st Signal/Mode	None
TR	13.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
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Sequence - Part 1

Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	T2-SPACE
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ABCD_T2w_SPC_vNav

TA: 6:35 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
TE	565.00 ms
Averages	1.0
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7;NC1,2

Contrast - Common

TR	3200.0 ms
TE	565.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T2 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	3200.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F30.0
L	0.0 mm
A	20.0 mm

Geometry - AutoAlign

F	30.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 A20.0 F30.0 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	3200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spc
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	241 Hz/Px
Echo Spacing	6.28 ms
Turbo Factor	200
Echo Train Duration	1187 ms

Sequence - Part 2

Introduction	On
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Sequence - Special

Include Nav.	On
Apply moco to	parent and nav
Remeasure	18 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Apply freq to	parent and nav

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\rfMRI REST ME PA XA60

TA: 7:04 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	60
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	1330.0 ms
TE 1	12.60 ms
TE 2	29.16 ms
TE 3	45.72 ms
TE 4	62.28 ms
Averages	1
Multi-band accel. factor	4
AutoAlign	Head > Brain

Contrast - Common

TR	1330.0 ms
TE 1	12.60 ms
TE 2	29.16 ms
TE 3	45.72 ms
TE 4	62.28 ms
MTC	Off
Magn. Preparation	None
Flip Angle	67 deg
Fat-Water Contrast	Fat Saturation
Contrasts	4
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	300

Contrast - Dynamic

Delay in TR	0.00 ms
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Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
Base Resolution	86
Phase Resolution	72 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	24
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	1330.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H6.0
L	0.0 mm
P	0.0 mm

Geometry - AutoAlign

H	6.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-16.0
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	150 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1330.0 ms
Log Signals	Off
Multi-band accel. factor	4

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	300
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	3230 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	62

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	3820 us
Min. prep scans	0
Min. prep scans SB	0
Inter-TE delay	0 us
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	On
Invert RO/PE polarity	On
Echoes in separate series	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\cmrr_mbep2d_diff

TA: 9:15 min Coil Selection: Auto Voxel Size: 2.3×2.3×2.3 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	221 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	5000.0 ms
TE	78.00 ms
Multi-band accel. factor	8
AutoAlign	---

Contrast - Common

TR	5000.0 ms
TE	78.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	221 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.3 mm
Base Resolution	96
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	221 mm
FOV Phase	100.0 %
Slice Thickness	2.3 mm
TR	5000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	221 mm
R >> L	221 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Multi-band accel. factor	8

Physio - PACE

Resp. Control	Off
Multi-band accel. factor	8

Diff

Diffusion Mode	Free
Diff. Directions	102
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	2000 s/mm ²
Averages	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Bandwidth	2264 Hz/Px
Echo Spacing	0.63 ms
Free Echo Spacing	Off
EPI Factor	96

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	2560 us
Refocus pulse duration	5120 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	Off
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\Investigators - validated on FIT\Frederick\Potpourri\ep2d_bold_mghTA: 8 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	223.0 ms
TE	98.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	223.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	223.0 ms
TE	98.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm

Geometry - Tim Planning Suite

Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	223.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On

BOLD

Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off
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Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
CC Mode	Direct adj coil
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ep2d_diff_mgh

TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
TE	200.00 ms
Concatenations	1
AutoAlign	---

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	500.0 ms
TE	200.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Contrast - Dynamic

Dynamic Mode	Standard
Delay in TR	0.00 ms

Geometry - Navigator**Resolution - Common**

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	500.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Diff

Diffusion Mode	Read
Diff. Directions	1
Diffusion Scheme	Bipolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
Optimization	None
EPI Factor	128

Sequence - Part 2

Introduction	Off
Phase Correction	Internal

Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ep2d_se_sms_mgh

TA: 1 sec Coil Selection: Auto Voxel Size: 3.9×3.9×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	285.0 ms
TE	182.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	285.0 ms
TE	182.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle 1	90 deg
Flip Angle 2	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	285.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	500 mm
R >> L	500 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	285.0 ms
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	752 Hz/Px
Echo Spacing	1.38 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off
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Sequence - Special

Imaging Dummy TRs	-1
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3
Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ABCD_fmri_rest_CMRR

TA: 5:14 min Coil Selection: Manual Voxel Size: 2.4×2.4×2.4 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	60
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.4 mm
TR	800.0 ms
TE	30.00 ms
Averages	1
Multi-band accel. factor	6
AutoAlign	---
Coil Elements	HC1-7

Contrast - Common

TR	800.0 ms
TE	30.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	52 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	383
Delay in TR	0.00 ms

Resolution - Common

FOV Read	216 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	2.4 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.4 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	6

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	383
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2778 Hz/Px
Echo Spacing	0.49 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\ABCD_fmri_rest_MGH

TA: 5:14 min Coil Selection: Auto Voxel Size: 2.4×2.4×2.4 mm³ Acc:: 12 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	60
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.4 mm
TR	800.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	800.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	383
Delay in TR	0.00 ms

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.4 mm
Base Resolution	90

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	44
SMS Factor	6
FOV Shift Factor	2
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	60
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	2.4 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	6 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	800.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	383
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2778 Hz/Px
Echo Spacing	0.49 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
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Sequence - Special

Imaging Dummy TRs	0
SMS ACS Dummy TRs	-1
FLEET Dummy Pulses	0
CC Mode	Direct adj coil
RF Clip	0
VERSE Factor	1.00
ACS mode	Standard
Kernel Size	3x3

Sequence - Special

Wait for TCP/IP Trigger	Off
Reverse Phase Encoding	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Potpourri\\can_neuromelanin

TA: 14 sec Coil Selection: Auto Voxel Size: 2.3×2.3×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
TE	10.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	100.0 ms
TE	10.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	25 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	100.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253263 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	100.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	25 deg
Measurements	1
Contrasts	1
TE	10.00 ms
TR	100.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Special

MT Flip Angle	370 degrees
MT Offset	1500 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s